Bayesian Machine Learning Project
When is Generalized Bayes Bayesian ? A Decision Theoretic Characterization of Loss-Based Updating

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Abstract

In the line of the paper When is Generalized Bayes Bayesian ? A Decision Theoretic Characterization of Loss-Based Updating by McAlinn and Takanashi (2026) we study the decision theoretic foundations of Gibbs learning. We present its practical implications and compare it to Bayes learning.

1 Machine learning from the decision theoretic point of view

Machine learning can be envisaged as a problem of decision under uncertainty. Given a partial observation of Nature in the form of data, the decision maker (DM) must choose a single machine in some set of possibilities. For this choice to be rational, the DM is required to rank his machines according to a certain criterion. Obvious criterions are performance on the train or test sets, but more real-world-relevant criterions are possible: absence of bias in the outputs of an LLM, computational resources required, scalability, etc…

The DM must now find a computable representation of these preference relations, so that the choice of a top ranking act reduces to a feasible optimization problem. However, nothing guarantees that such a representation exists. Therefore, the DM needs to constrain himself to a particular class of representable preferences, by imposing certain axioms. This is where decision theory becomes relevant.

1.1 Bayes learning and decision theory

A convenient class of optimization problems are those defined by Subjective Expected utility (SEU). The set of SEU-representable preferences is defined by the axioms of Savage decision theory [1]. An important feature of SEU is that it naturally gives rise to Bayes learning as a rational and dynamically consistent decision method. In ML, the states of the DM typically consist of an observed part, containing training data, and an unobserved part containing future predictions and/or model parameters. By letting the DM specify a joint probability on states, Savage theory naturally induces the belief update interpretation of the posterior.

Another approach to SEU has been formalized by Anscombe and Aumann, recalled in [1]. This theory differs from Savage's in a few ways, especially in its axioms over preferences: weak order, independence under mixtures, Archimedean property, monotonicity and non-triviality. However, it leads to the same SEU representation and to Bayes rule. For our project, its main interest is that it can be generalized to induce a broader class of preference representations that will in turn imply Gibbs learning.

1.2 Misspecification phenomenons and averaged predictors

Bayes learning, despite its theoretic generality, is not without important drawbacks. First, the likelihood may sometimes be untractable or undefined, leading to alternatives such as simulation-based inference. Moreover, Bayesian models can suffer from misspecification issues as illustrated by the following toy experiment.

Consider the problem of estimating the mean of an unbalanced gaussian mixture with true mean 0:

$$Y \sim \alpha\mathcal{N}(0,1) + (1-\alpha)\mathcal{N}(0,\sigma^2) \tag{1}$$

where α is close to 1 and $\sigma^2 \gg 1$. Among our simulations, we test three different misspecified estimators: a standard Bayes estimator, a tempered version of this estimator and a Gibbs model:

$$\begin{aligned} \text{Standard Bayes: } & p(\theta|y) \propto p(y|\theta)p(\theta) \\ \text{Tempered Bayes: } & p(\theta|y) \propto p(y|\theta)^\eta p(\theta) \\ \text{Gibbs: } & q(\theta|y) \propto \exp(-\eta l(\theta, y)) \end{aligned} \tag{2}$$

with $p(y|\theta)$ a gaussian likelihood, l the Huber loss and π a normal prior on the mean θ.

Our results show that while the standard Bayes model is very sensitive to outliers, especially in a few data regime, the tempered Gibbs models achieve more accurate estimations. In particular, the Gibbs model possesses the ability to use an adapted loss function, while the tempered model achieves better accuracy at the cost of non-concentration.

Similar phenomenon are described in [2] for simple prediction tasks. Indeed, when one solves the prediction problem using Savage theory and log-loss as utility, one obtains an averaged predictor, weighted by the model's posterior. This phenomenon, illustrated in [2], can be seen as an extension of the model space to its convex hull. Moreover, their authors observe that models that originally concentrate, can acquire a mixture behavior through exponential tempering, potentially leading to improved performances.

These examples illustrate how Gibbs learning and exponential tempering can be very interesting in practice. The difficulty, from a decision theoretic viewpoint, is to understand where Gibbs learning comes from. Indeed, the form of Gibbs posteriors are not implied by Savage or Anscombe-Aumann theories of SEU, making the usual belief posterior interpretation inexact.

2 Variational preferences

It turns out their exists a natural way of explaining Gibbs posteriors from a decision theoretic setting. By weakening certain axioms of Anscombe-Aumann theory, authors in [3] have derived a set of axioms that naturally lead to variational representations of preferences.

2.1 Axioms of variational representation

By weakening the independence axiom, Theorem 13 from [3] asserts that there exists a distribution π over states, a loss l, unique up to affine transformations, and c a convex, lower-semicontinuous function from distributions over states to $\mathbb{R}$, such that:

$$f \succsim g \Leftrightarrow \min_{q \ll \pi} \int l(f)dq + c(q) \leq \min_{q \ll \pi} \int l(g)dq + c(q) \tag{3}$$

In particular, this authorizes the use of divergences $c(q) = \frac{1}{\eta}D(q, \pi)$. From now on, we will refer to this representation as Variational Expected Utility (VEU). Note that $q \ll \pi$ denotes absolute continuity.

To complete the picture, we can add axiom A.4 from [4]. This separability axiom assumes that if states factor into distinct parts $\Theta = \Theta_1 \times \Theta_2$, with $\pi = \pi_1 \otimes \pi_2$ factoring independently across these parts, and $l(\theta) = l_1(\theta_1) + l_2(\theta_2)$ is additive over them, then q must factor into $q_1 \otimes q_2$. In other words, two independent problems have independent decision rules. This axiom forces c to be the KL-divergence [4].

2.2 An example: prediction via variational representations

To illustrate how this representation leads to Gibbs methods, we derive its application to prediction problems. We define our states as $\mathcal{S} = \mathcal{X}^n \times \mathcal{Y}^n \times \Theta$ and consequences $\mathcal{C} = \{\mathcal{X} \to \mathcal{Y}\}$. We restrict ourselves to a set of acts $\mathcal{A}$ that do not depend on Θ and map into degenerate distributions over consequences: an act f maps training data to a predictor function. We choose π, l and η. The VEU representation defines the best act as:

$$a^*_{x_{\mathrm{tr}}, y_{\mathrm{tr}}} = \arg\min_{a \in \mathcal{A}} \min_{q \ll \pi} \int_\Theta l(a, \theta, x_{\mathrm{tr}}, y_{\mathrm{tr}}) dq(\theta) + \frac{1}{\eta} \mathrm{KL}(q, \pi) \tag{4}$$

The solution to the first optimization problem, as proved in [4], is the Gibbs posterior:

$$q^*(\theta) = q(\theta | x_{\mathrm{tr}}, y_{\mathrm{tr}}) = \frac{1}{Z} \exp(-\eta l(a, \theta, x_{\mathrm{tr}}, y_{\mathrm{tr}})) \pi(\theta) \tag{5}$$

where $Z = \int_\theta \exp(-\eta l(a, \theta, x_{\mathrm{tr}}, y_{\mathrm{tr}})) \pi(\theta) d\theta$.

The optimization over acts then reduces to an SEU:

$$a^*_{x_{\mathrm{tr}}, y_{\mathrm{tr}}} = \arg\min_{a \in \mathcal{A}} \int_\Theta l(a, \theta, x_{\mathrm{tr}}, y_{\mathrm{tr}}) dq^*(\theta) \tag{6}$$

Now, if $l(a, \theta, x_{\mathrm{tr}}, y_{\mathrm{tr}}) = \sum_i \|y_i - f_\theta(x_i)\|^2$ with (f_θ) a family of parametrized models (e.g. neural networks) and by identifying $y_i = a_{x_{\mathrm{tr}}, y_{\mathrm{tr}}}(x_i)$, we obtain:

$$a^*_{x_{\mathrm{tr}}, y_{\mathrm{tr}}} = \arg\min_{a \in \mathcal{A}} \sum_i \int_\Theta \left\| a_{x_{\mathrm{tr}}, y_{\mathrm{tr}}}(x_i) - f_\theta(x_i) \right\|^2 dq^*(\theta) \tag{7}$$

We minimize the interior integral for each x_i. By using bias-variance decomposition:

$$\begin{aligned} a^*_{x_{\mathrm{tr}}, y_{\mathrm{tr}}}(x_i) &= \arg\min_{a \in \mathcal{A}} \left\| a_{x_{\mathrm{tr}}, y_{\mathrm{tr}}}(x_i) - \mathbb{E}_{\theta \sim q^*}[f_\theta(x_i)] \right\|^2 + \mathrm{Var}_{\theta \sim q}(f_\theta(x_i)) \\ &= \mathbb{E}_{\theta \sim q^*}[f_\theta(x_i)] \end{aligned} \tag{8}$$

We obtain the averaged predictor over θ.

We note a slight issue however: according to the theory, π should be a distribution on the entire state space, not just over Θ. We believe that this is not a fundamental flaw of the theory, but rather a problem in our definition of states and acts. By slightly tweaking the setup or adding restrictions, it might be possible to separate π from the other state variables.

2.3 SEU as a special case of VEU

A very important property of VEU is that it contains SEU as a special case. Proposition 22 from [3] implies that:

$$\min_{q \ll \pi} \int l(f) dq + \frac{1}{\eta} \mathrm{KL}(q, \pi) \underset{\eta \to \infty}{\longrightarrow} \int l(f) dq \tag{9}$$

Therefore, Bayes learning corresponds to a limiting case of VEU with $\eta = \infty$. Another way to see this is to note that $\exp(-\eta l(\theta)) \underset{\eta \to \infty}{\longrightarrow} \mathbb{1}_{\arg\min l(\theta)}$ if $\min_\theta l(\theta) = 0$ (which can be achieved by anchoring the loss). This is why Proposition 5 in [4] states that Gibbs posteriors become degenerate under the von Neumann-Morgenstern axioms (which are the foundations of linear utility). We will see in Section 3.2 that $\eta = \infty$ corresponds to a full confidence regime.

Moreover, Theorem 1 from [4] shows that VEU and SEU also cross paths when the loss is negative log-likelihood and $\eta = 1$, which corresponds to the prediction regime with log-loss.

3 Interpretations of VEU

3.1 The nature of Gibbs updates

As the authors [4] stress, a Gibbs posterior is not a belief posterior. The distribution q^* in Section 2.2 does not arise from a Bayes rule update. Actually, it is more accurate to view q^* as an estimation act. The notation $q^*(\theta) = q(\theta|x_{\text{tr}}, y_{\text{tr}})$ only hints at the dependence of q^* on observed data, not at a Bayes rule application. Hence, the vocabulary of Bayesian analysis does not, in general, carry over to Gibbs learning. As we will see in the next section, q^* more accurately reflects an optimal mixture of risk VS confidence management. For these reasons, we prefer to refer to q^* as a decision posterior rather than a belief posterior.

3.2 Risk VS confidence

When a DM faces the problem of choosing a machine rationally, they are confronted with two sources of uncertainty.

The first of these sources is called risk. Risk represents the inevitable randomness of real-world phenomena. If the DM already knows the exact class of machines that represent the phenomena, all that remains is to explain the noise in the data. For instance, if the DM knows exactly that his data is driven by a linear mechanism (this certitude might come e.g. from a law of physics), risk can be represented by additive gaussian noise: $Y = AX + \varepsilon$. The go-to Bayesian way is to choose a gaussian likelihood $Y \sim \mathcal{N}(AX, \sigma^2)$ and to encode risk into the prior π as belief. The Bayes update then follows the evolution of this belief, until residuals become normally distributed.

The second source of risk is confidence. It reflects how much the DM believes in their choice of model class. In many important applications, such as NLP, it is standard to assume that the DM's model class is obviously wrong, due to an immense lack of knowledge on the phenomenon being observed. This doesn't mean that the machines contained in the class are irrelevant though: they can still bring interesting solutions w.r.t. the loss chosen by the user. This is what authors in [2] call "obviously-wrong-but-pretty-adequate" models. By minimizing $\text{KL}(q, \pi)$ the model takes into account the DM's lack of confidence, represented by the parameter η. As we have seen in Section 2.3, Bayes learning corresponds to a complete confidence regime, where the DM is certain to have chosen the right model class.

3.3 Maximum entropy, PAC-Bayes bounds and SGD

VEU makes several connections with other concepts from across the ML landscape. The connection with PAC-Bayes bounds is immediate, as the quantity being minimized in VEU is exactly the MacAllester bound on generalization error.

In the same line, we can link this phenomenon to the limit behavior of SGD [5]. It can be shown that, in the continuous limit, SGD defines a Langevin SDE on model parameters θ:

$$\dot{\theta}_t = -\nabla l(\theta)dt + \sqrt{2\eta}dW_t \tag{10}$$

with W_t a Brownian motion and η depending on the batch size. It turns out that the limiting distribution of this system, given by the Fokker-Planck equation, is:

$$p(\theta) \propto \frac{1}{Z}\exp\left(-\frac{1}{\eta}l(\theta)\right) \tag{11}$$

SGD thus meets VEU in the limit of continuous updating. In this case, the model's confidence comes from the batch size. SGD does not commit to sharp minima if the next batch has a probability of being different from the current one: the smaller the batches, the greater the uncertainty. With a single batch, the update becomes deterministic, and is prone to overfitting exactly like Bayesian models.

VEU, PAC-Bayes bounds and SGD limit behavior all point towards the max entropy principle. This principle, tightly linked to Occam's razor, states that a machine should put minimal effort into fitting the constraints imposed by data, while keeping maximum entropy over its remaining parameters, something that over-parametrized neural networks typically do [6]. From this point of view, averaged predictors and non-concentration can be seen as an attempt of the model to "over-parametrize" itself and maximize its entropy.

4 Implications for Gibbs learning

4.1 The choice of η and π

We restate the foundational claims of the authors of [4] in the light of VEU representations.

First, η is a subjective parameter, as much as the loss l and the prior π. Therefore, it is not a parameter to be optimized according to l, it is a defining feature of the preference relation that the DM represents. Consequently, it is as important to describe the choice of η as much as the choice of the prior and the choice of loss. For example, authors in [2] select η using the SafeBayes algorithm.

Second, π does not play the role of a prior belief. To quote [4], π is the "default rule" that we stick to when we have very low confidence in our model. Even when confidence is higher, π still acts as a regularization on the learning process. Hence, a good choice of π is capital, especially if one wishes to benefit from maximum entropy. However, π can still encode strong real-world assumptions on the data e.g. by being restricted on a compact set or on continuous functions.

4.2 Model comparison

Another important implication of VEU lies in model comparisons. When a Bayesian specifies a joint model over states, they specify a model of Nature. Comparing two Bayesian models amounts to looking at which model is more favored by the data. This naturally leads to Bayes factors analysis, through the ratio of marginal likelihoods.

In Gibbs learning, this comparison procedure falls apart for two reasons. First, the Gibbs DM does not specify any joint model over states: Gibbs machines are not models of Nature, but (η)-cautious attempts at achieving a certain objective: the partition functions Z do not inherit a Bayes factor interpretation. Additionally, in the particular case of log-loss and $\eta \neq 1$, it is important to stress that the likelihood does not define a model of Nature anymore, but an objective to optimize.

Second, VEU defines the loss l uniquely only up to affine transformations. A simple calculation in [4] shows that changing l to $l' = l + r$, where r is independent of θ, induces the same posterior q^* (thanks to normalization) but changes the partition function to $Z' = \exp(-\eta r)Z$, cancelling its interpretability. [4] recommends using the prequential score, which is invariant to such transformations.

4.3 The role of KL

The use of KL-divergence in many applications should not be taken as canonical. Recall in Section 2.1, that the use of KL is constrained only if one abides by the separability axiom A.4 of [4]. Even if most problems do abide by it, other settings authorize many other divergences, as the VEU representation theorem of [3] admits a very general form.

4.4 The prediction method

As seen during the derivation of Section 2.2, the averaged predictor is obtained as a minimizer of a specific optimization problem, using $l(a, \theta, x_{\text{tr}}, y_{\text{tr}}) = \sum_i \|y_i - f_\theta(x_i)\|^2$. If one chooses a different loss, one gets a different problem. For example, with the δ-Huber loss, we obtain at the end:

$$\begin{aligned}\arg\min_{a\in\mathcal{A}} \frac{1}{2}\left\|a_{x_{\text{tr}},y_{\text{tr}}}(x_i) - \mathbb{E}_{\theta\sim q^*}[f_\theta(x_i)]\right\|^2 \mathbb{1}_{\left|a_{x_{\text{tr}},y_{\text{tr}}}(x_i)-f_\theta(x_i)\right|\leq\delta} \\ +\delta\mathbb{E}_{\theta\sim q^*}\left[\left|a_{x_{\text{tr}},y_{\text{tr}}}(x_i) - f_\theta(x_i)\right|\right]\mathbb{1}_{\left|a_{x_{\text{tr}},y_{\text{tr}}}(x_i)-f_\theta(x_i)\right|>\delta}\end{aligned} \tag{12}$$

This problem, if it even admits a solution, might lead to a different predictor, possibly a mixture of averaged and median predictors.

Hence, applications relying on averaged predictors are VEU-rational only if they use the gaussian log likelihood as a loss. However, these methods can still go beyond Bayesian learning by tuning the parameter η, as in [2].

5 Discussion

We suspect that more connections can be made between VEU and other domains of ML and mathematics. It might be interesting to put VEU in regards of Singular Learning Theory [7], a domain of ML that seeks to analyze models with non-positive-definite Fischer information matrix, borrowing techniques from complex analysis and algebraic geometry. This could provide a generalization of the convex hull extension phenomenon to arbitrary Gibbs models, through Hironaka's resolution of singularities. We suspect that another connection could be made with the field of Non-Commutative Geometry [8], pioneered mainly for theoretical quantum physics, leading to objects that are reminiscent of averaged predictors . A link might also be made with causal inference [9], precisely via the latter link, if it proves to be true. Finally, it would be interesting to explore more deeply the implications of VEU in the nonparametric setting.

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